Project Instructions

Operational research for urban solar development

"Regional PV production estimation from open data"

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Project summary

The project aims at estimating the hourly photovoltaic (PV) production of a whole French region from open data, and at explaining the gap between your estimate and the production published by the grid operator.

The first objective, individual, is the estimation of the operating variables of one PV system: plane-of-array (POA) irradiance, module temperature, DC power and AC power, from weather data. You compare each estimate with the operating data of the system.

The second objective, in pairs, is the estimation of the production of the whole PV fleet of a region, built from the national register of installations and compared with the hourly series of RTE. You then attribute the gap to a named cause and test your hypothesis.

Every student receives a student number **n** for part A. Every pair receives a pair number **k** and one region for part B. The file `pairs.csv` gives your numbers and your region.

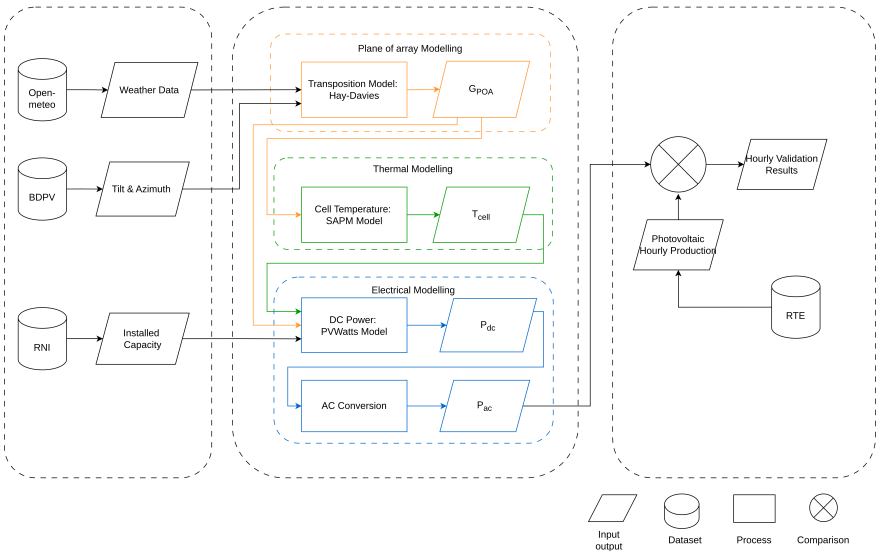


Figure 1. The simulation chain of the project. Weather data, orientations and installed capacity feed the POA, thermal and electrical models, and the result is compared with the RTE series. Ennhiri et al., EU PVSEC 2026.

Deadline for the notebook and the csv files: Tuesday 01/12/2026 at 23h59.

Oral presentation of the pair work: Wednesday 02/12/2026, afternoon, both TP groups together.

- If late, less than 15 min: −10 %.
- If late, less than 1 h: −50 %.
- If late, more than 1 h: −100 %.

Deliverables

- One Jupyter notebook per pair, exported as an html file, running from top to bottom without error. The notebook holds part A for each member and part B for the pair.
- csv files with a DateTime index in UTC covering the year 2023, named with your numbers:
  - `student_n_acestimation.csv`, one per member, for part A.
  - `pair_k_region.csv`, for part B, the hourly production of your region.
  - `pair_k_classes.csv`, for part B, your representation of the fleet.
- One oral presentation per pair on part B, 10 minutes plus questions, sent as pdf the evening before.

Part A is graded on the notebook and the csv files only. The oral presentation covers part B only.

Send all files in one zip file to **mohamed.ennhiri@univ-smb.fr** before the deadline, and make sure the mail is received. Export a `pd.DataFrame` with the `to_csv()` command.

Input datasets of the project

The project notebook loads the datasets from the course repository according to your numbers.

Part A, per student n, in `project/data/`

- `student_n_inputs.csv`: hourly weather at the site of your system, 2023, UTC. Columns `ghi`, `dhi`, `dni` in $\mathrm{W\,m^{-2}}$, `temp_air` in °C, `wind_speed` in $\mathrm{m\,s^{-1}}$ at 10 m, `solar_azimuth` and `solar_elevation` in degrees. Source: PVGIS SARAH3.
- `student_n_actuals.csv`: the operating data of your system, to compare with the outputs of your models. Columns `g_poa` in $\mathrm{W\,m^{-2}}$, `t_mod` in °C, `p_dc` and `p_ac` in W.
- `student_n_system.csv`: site, department, region, latitude, longitude, tilt, azimuth, nameplate DC power in kWp, inverter AC rating in kW, albedo.

Part B, per region, in `project/regions/<region>/`

- `register.csv`: every PV installation of the region in the national register at 31/12/2023, with department, commune, capacity in kW, commissioning date, latitude and longitude. The register groups small rooftop systems per commune. Source: ODRE.

- `nodes.csv`: one weather node per department, the commune closest to the capacity-weighted centre of the department, with the capacity of the department.
- `weather_dept_<dd>.csv`: hourly weather at the node of each department, same columns as part A.
- `reference_2023.csv`: hourly PV production of the region in MW, 2023, UTC, the hourly mean of the half-hour values of RTE eCO2mix. Corse sits outside the RTE grid: its series comes from the hourly production published by EDF SEI.

PV installation data, part A

Read your values in `student_n_system.csv`. Azimuth follows the pvlib convention: clockwise from north, south at 180°. Modules are crystalline silicon. The datasheet of your modules is unknown: choose a temperature coefficient of power between -0.29 and -0.45 $\%\,\mathrm{K}^{-1}$ and state your choice. Albedo 0.2.

Tasks

A. AC power modelling of your system, individual

1. Estimate the POA irradiance.
2. Estimate the module temperature.
3. Estimate the DC power.
4. Estimate the AC power.
5. Calculate the performance of the installation at the AC level over the whole year through two indicators:
   (a) the performance ratio.
   (b) the energy performance index, measured energy divided by modelled energy.

Compare each estimate with your actuals: one plot of one week, one scatter plot, MBE and RMSE over the year. Notebook 3 shows the method on the course site.

Expected outputs

- `student_n_acestimation.csv` with four columns, `gpoa_estimated`, `t_mod_estimated`, `dc_power_estimated`, `ac_power_estimated`, in $\mathrm{W\,m^{-2}}$, °C, W and W, with a DateTime index.
- The notebook with your method, the comparison plots, and the PR and EPI printed.

B. Production of your regional fleet, in pairs

0. **Data check.** Rebuild two numbers from the raw open data and compare them with the provided files. First, the annual PV energy of your region in 2023 from the eCO2mix regional dataset. Second, the installed capacity of your region from the national register, keeping the installations commissioned before 01/01/2024. State the gap and its cause. Three traps to check: the time zone, the half-hour time step, and the capacity column of the register.

1. **Installed capacity.** Capacity by department, by size class (below 9, 9 to 36, 36 to 100, 100 to 250, above 250 kWp), and cumulative capacity as a function of the commissioning date.

2. **A first model.** Every system at 30°, facing south, one weather node per department, the capacity-weighted sum over the departments. Compare with RTE on one week and on the mean daily profile. Write what the profile reveals about your assumption.

3. **A better representation of the fleet.** Build classes with a tilt, an azimuth and a share, from what the comparison shows. State the reasoning or the source behind the shares, and the change in the metrics.

4. **Comparison with RTE.** MBE, RMSE, nRMSE with its denominator, and R^2, at hourly, daily and monthly scale, on the same hours. One week, the mean daily profile and the monthly energy, plotted against RTE.

5. **Attribution.** Split the gap between installed capacity and physics by dividing both series by their annual energy. Read the pattern of the residual, test one hypothesis by changing one input, and report the effect on the metrics and the gap left unexplained.

6. **Bonus.** Run your region with one weather node instead of one per department, and report the change in the metrics.

Raw data for task B.0

- eCO2mix regional: `https://odre.opendatasoft.com/explore/dataset/eco2mix-regional-cons-def/`
- National register: `https://odre.opendatasoft.com/explore/dataset/registre-national-installation-production-stockage-electricite-agrege/`
- Corse, EDF SEI: `https://opendata-corse.edf.fr`, dataset "Production d'électricité par filière et coûts de production au pas horaire".

The online register is a recent snapshot. Systems removed since 2023 are absent from the online version, so expect a small gap with the provided file. Explain the gap, do not hide the gap.

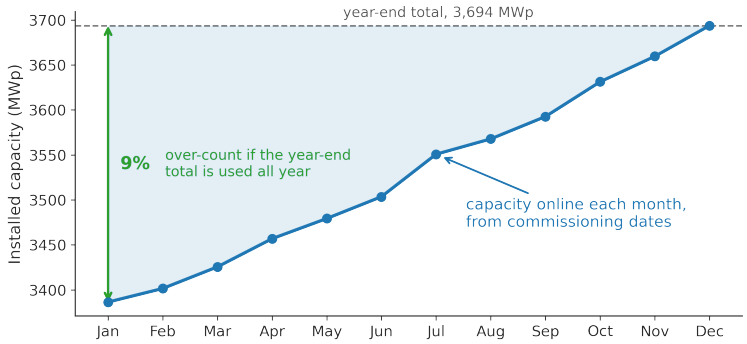


Figure 2. Task B.1. Installed capacity of Occitanie in each month of 2023, from the commissioning dates of the register. Using the year-end total for the whole year over-counts the capacity by 9 % in January. Ennhiri et al., EU PVSEC 2026.

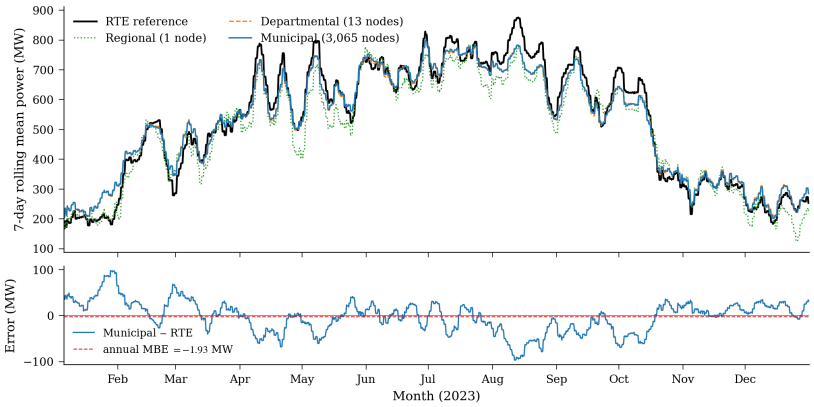


Figure 3. Task B.4. Seven-day rolling mean of the modelled and RTE production for Occitanie, 2023, and the error of the model. Your comparison looks like this one, for your region. Ennhiri et al., EU PVSEC 2026.

Expected outputs

- `pair_k_region.csv`: one column, `p_region_mw`, hourly, 2023, DateTime index in UTC.
- `pair_k_classes.csv`: one row per class, columns `tilt`, `azimuth`, `share`.
- The notebook with the plots, the metrics table at three time scales, the finding of task B.2 written in a text cell, and the attribution of task B.5.

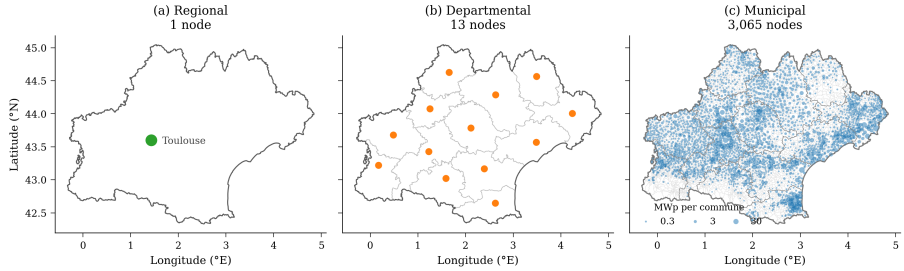


Figure 4. Bonus task. The same fleet simulated with one node for the region, one node per department, and one node per commune. Ennhiri et al., EU PVSEC 2026.

Grading scale

Score distribution: 80 % notebook and csv files, 20 % oral presentation. Part A is graded per member, from the notebook and the csv files only. Part B is graded per pair, from the notebook, the csv files and the oral presentation.

- **30 % A. AC power modelling, individual.** 6 % for each of the five tasks. For tasks 1 to 4: relevance of the estimation method, and RMSE accuracy of the estimation, 3/5 for the accuracy of the most basic method, 5/5 for the accuracy of the teacher's method. For task 5: correctness of the calculation.
- **45 % B. Regional fleet, pair.** 5 % for task B.0, then 8 % for each of tasks B.1 to B.5. Relevance of the approach, accuracy against RTE, the finding of task B.2 written from the profile, the reasoning behind your classes, and one hypothesis tested with its effect measured.

- **5 %** overall structure and ease of reviewing the notebook.
- **20 % oral presentation of part B, pair.** 10 minutes per pair, then questions to each member. Suggested structure:

  1. Your region: installed capacity, size classes, growth during 2023, the data check of task B.0.
  2. The first model, and what the mean daily profile revealed.
  3. Your representation of the fleet: classes, shares, and the reasoning behind them.
  4. Comparison with RTE: one week, mean daily profile, monthly energy, metrics with their denominators.
  5. Attribution: the split, the hypothesis tested, the gap left unexplained.

  Points: 8 content, 4 figures with axes and units, 4 timing and clarity, 4 answers to questions.
- **Bonus 10 %**: task B.6.

Potential bonus or penalty points: original approaches, better accuracy than the teacher's benchmark, figures explaining the approach and the results, a number without a unit.

Advice

- Refer to the course formulas and to the pvlib documentation.
- Write short explanations. A stated limitation counts as a result.
- Show your approach with relevant plots: one plot per result, axes labelled with units.
- Comment your code with `#`. For example:

```
# PVGIS labels each hour at HH:10, RTE at HH:00: same labels before any comparison
weather.index = weather.index.floor("h")

# Keep the hours present in both the model and the reference
both = pd.concat([p_region, reference["ref_solar_mw"]], axis=1, join="inner")
```

Acknowledgement and reference

The structure of this project, its deliverables and its grading scale follow the project of the course given by Alexandre Mathieu at the Solar Academy in 2024, "PV failure detection based on operational time series".

Figures: M. Ennhiri, A. Boccalatte, S. Plassart, B. Nérot, M. Thebault, "Impact of the spatial aggregation level on the estimation of distributed PV production: a case study in Occitanie, France", EU PVSEC 2026. Data: pvlib, PVGIS SARAH3, ODRE, RTE eCO2mix, EDF SEI.